HW 1

ENGR 21, Fall 2026.

2026-09-10

Due Date	Thu, Sep 10
Turn in link	Gradescope
URL	`emadmasroor.github.io/E21-F26/Homework/HW1`

What to turn in

For this assignment, turn in a single PDF file, even though you will be writing code for some parts. Make sure your code is legible, indented, and syntax-highlighted; a screenshot of code is usually a bad idea, but OK for this homework.

You may use the text editor in Thonny Code Editor and then go to File > Print. If you use this option to make a printed copy of code for submission, please increase the font size using the 'scale' option when printing.

1 Writing simple programs for the Circuit Playground Express

Using only the features of Python that have been discussed so far,

1. Write a program for the Circuit Playground Express that plays a tone of 440 Hz for one-second intervals repeatedly until a cell phone light is shined on it, at which point the program should terminate.

2. Write a program for the Circuit Playground Express that plays a tone of 440 Hz for one-second intervals repeatedly. When it is brought close to a warm object (warmer than room temperature), it stops playing this tone and instead plays a high-frequency tone of 880 Hz for a half second. After this, the program should terminate.

2 Truth Tables

We have seen that the truth table for the `and` and `or` operators was as given in lecture.

> **Note**
>
> It is possible to use logic to determine the truth tables for more complicated operators than `and` and `or`, but here you should simply check all possibilities using Python.

2.1 For a binary operator

Use this info to construct a truth table for the following binary operator.

```
(a and b) or a
```

> **Note**
>
> To do this, make a table with the following entries.

Variable `a`	Variable `b`	Result of `(a and b) or a`
`True`	`True`	
`True`	`False`	
`False`	`True`	
`False`	`False`	

2.2 For an operator with three inputs

Construct a truth table for the following operator.

```
(a or b) and c
```

> **Note**
>
> One row has been filled out for you.

`a`	`b`	`c`	`(a or b) and c`
`True`	`True`	`False`	`False`

> **How many rows do you need?**
>
> The rows of a truth table should cover every possible combination of True and False values. If binary operators required 4 rows, how many rows do you need for a logical operator with three inputs, such as `(a or b) and c`?

3 While loops

Determine the behavior of the Circuit Playground Express when the following code is run on it. Give your answer in words.

3.1 No inputs

1. ```
   while True:
     cpx.play_tone(440,1)
   cpx.play_tone(880,0.5)
   ```

2. ```
   while True:
     cpx.play_tone(880,0.5)
   cpx.play_tone(440,1)
   ```

3. ```
   while True:
     cpx.play_tone(880,0.5)
     cpx.play_tone(440,1)
   ```

4. ```
   while True:
     cpx.play_tone(440,1)
     cpx.play_tone(880,0.5)
   ```

5. ```
   cpx.play_tone(440,1)
   while True:
     cpx.play_tone(880,0.5)
   ```

6. ```
   cpx.play_tone(440,1)
   while False:
     cpx.play_tone(880,0.5)
   ```

7. ```
   cpx.play_tone(440,1)
   while False:
     cpx.play_tone(880,0.5)
   cpx.play_tone(440,1)
   ```

3.2 With inputs

For the following questions, determine what the board will do under the different relevant possibilities. For example,

- if a flashlight is shining on it vs. not
- if the temperature sensor is above/below a threshold
- if a certain pin is touched or not touched.
   ```

1.
```
while not (cpx.touch_A1 and cpx.touch_A5):
  cpx.play_tone(440,1)
cpx.play_tone(880,0.5)
```

2.
```
while not (cpx.touch_A1 and cpx.temperature > 30):
  cpx.play_tone(440,1)
cpx.play_tone(880,0.5)
```

3.
```
while not (cpx.touch_A1 or cpx.temperature > 30):
  cpx.play_tone(440,1)
cpx.play_tone(880,0.5)
```

4.
```
cpx.play_tone(440,1)
while cpx.light < 100:
  cpx.play_tone(880,0.5)
cpx.play_tone(440,1)
```

4 Nested while loops

1. Explain the difference between the following programs.

```
while True:
    cpx.play_tone(880,0.5)
    while True:
        cpx.play_tone(440,1)
```

```
while True:
    cpx.play_tone(880,0.5)
while True:
    cpx.play_tone(440,1)
```

2. For each of the following parts, assume that the Circuit Playground Express has been running the following code for a few seconds before you carry out your intervention. (i.e., a, b, c and d are **not** applied sequentially.)

```
while not cpx.touch_A7:
 cpx.play_tone(880,0.5)
 while not cpx.touch_A5:
     cpx.play_tone(440,1)
```

a. What happens if pin A5 is tapped ?
b. What happens if pin A5 is touched for ten seconds ? Then, when you release it, what happens?
c. What happens if pin A7 is touched while the speaker is beeping at 440 Hz ?
d. What happens if both pin A5 and A7 are pressed for a few seconds?

3. If the Circuit Playground Express has been running the following code for a few seconds, what happens if pin A5 is touched?

```
while not cpx.touch_A7:
   cpx.play_tone(880,0.5)
   while not cpx.touch_A5:
       cpx.play_tone(440,1)
   while True:
       cpx.play_tone(220,1)
```